

Yanfang Liu

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EDUCATION

The Chinese University of Hong Kong (CUHK), Hong Kong 07/2025 — 06/2029
Ph.D., Computer Science and Engineering
Research Topic: Design Space Exploration, Statistical Modeling.

Beihang University, China 09/2022 — 01/2025
M.Sc., Electronic Information GPA: 3.80/4.00 (Rank: 1/59)
Research Topic: Statistical Modeling, Design Space Exploration.

University of Jinan, China 09/2018 — 06/2022
B.Eng., Electrical Engineering and Automation GPA: 4.66/5.00 (Rank: 2/179)

Publications

- [1] **Yanfang Liu**, Peng Xu, Yapeng Li, Mingjun Wang, Leilei Jin, Rongliang Fu, Tinghuan Chen, Wei Xing, Bei Yu, and Tsung-Yi Ho. “MigSizer: Physical-aware Sizing Improving Analog/RF Circuit Migration”. In: *IEEE/ACM International Conference on Computer Design (ICCD)*. Hong Kong, Nov. 16-18, 2026.
- [2] **Yanfang Liu**, Zijin Cheng, Mingjun Wang, Rongliang Fu, Chao Wang, and Bei Yu. “LibPilot: Knowledge-Constrained Dual-Agent Reasoning for Standard Cell Library Tuning”. In: *International Conference on Computer Aided Design (ICCAD)*. San Jose, CA, USA, Nov. 08-12, 2026.
- [3] **Yanfang Liu**, Mingjun Wang, Peng Xu, Rongliang Fu, Bei Yu, and Tsung-Yi. Ho. “AnalogVerifier: A Neuro-Symbolic Framework for Analog Circuit Verification”. In: *International Conference on Machine Learning (ICML)*. Seoul, South Korea, Jul. 6–11, 2026.
- [4] **Yanfang Liu**, Guohao Dai, and Wei W. Xing. “Seeking the Yield Barrier: High-Dimensional SRAM Evaluation Through Optimal Manifold”. In: *2023 60th ACM/IEEE Design Automation Conference (DAC)*. 2023, pp. 1–6. DOI: 10.1109/DAC56929.2023.10247952.
- [5] **Yanfang Liu**, Guohao Dai, Yuanqing Cheng, Wang Kang, and Wei W. Xing. “OPT: Optimal Proposal Transfer for Efficient Yield Optimization for Analog and SRAM Circuits”. In: *2023 IEEE/ACM International Conference on Computer Aided Design (ICCAD)*. (**Best Paper Nomination**). 2023, pp. 1–9. DOI: 10.1109/ICCAD57390.2023.10323689.
- [6] **Yanfang Liu**, Lei He, and Wei W. Xing. “Beyond the Yield Barrier: Variational Importance Sampling Yield Analysis”. In: *Proceedings of the 43rd IEEE/ACM International Conference on Computer-Aided Design*. ICCAD ’24. Newark Liberty International Airport Marriott, New York, NY, USA: Association for Computing Machinery, 2025. DOI: 10.1145/3676536.3676672.
- [7] **Yanfang Liu** and Wei W. Xing. “FUSIS: Fusing Surrogate Models and Importance Sampling for Efficient Yield Estimation”. In: *2025 Design, Automation & Test in Europe Conference (DATE)*. 2025, pp. 1–7. DOI: 10.23919/DATE64628.2025.10993100.

- [8] **Yanfang Liu** and Wei W. Xing. “CIS: Conditional Importance Sampling for Yield Optimization of Analog and SRAM Circuits”. In: *2024 29th Asia and South Pacific Design Automation Conference (ASP-DAC)*. 2024, pp. 386–391. DOI: 10.1109/ASP-DAC58780.2024.10473819.
- [9] Wei W. Xing, **Yanfang Liu**, Weijian Fan, and Lei He. “Every Failure Is A Lesson: Utilizing All Failure Samples To Deliver Tuning-Free Efficient Yield Evaluation”. In: *Proceedings of the 61st ACM/IEEE Design Automation Conference*. DAC ’24. San Francisco, CA, USA, 2024. DOI: 10.1145/3649329.3657381.
- [10] Zhongru Xiong, Mingjun Wang, **Yanfang Liu**, Tsung-Yi Ho, Bei Yu, and Binwu Zhu. “HOIST: A High-Order Interaction Surrogate for Sample-Efficient High-Sigma Yield Estimation in High-Dimensional Circuits”. In: *2027 32th IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC)*. okyo, Japan, Jan. 25-28, 2027.
- [11] Shiyuan He, Shunxiang Yu, Luhao Wang, **Yanfang Liu**, Xiuhan Lin, and Wenjing Han. “Trade-off Stochastic Game Based Bidding Strategy for Multiple Wind Farms”. In: *2021 IEEE/IAS Industrial and Commercial Power System Asia (I&CPS Asia)*. 2021, pp. 1398–1403. DOI: 10.1109/ICPSAsia52756.2021.9621692.

AWARDS

- **Beijing Outstanding Graduate** 2025
- **Top 10 Graduate Scholarship** (The highest honor in Beihang University) 2024
- **National Scholarship** (Master) 2024
- **National Scholarship** (Master) 2023
- **Song Jian Scholarship** (The highest honor in UJN) 2022
- **National Scholarship** (Bachelor) 2020
- **2023 ICCAD Best Paper Nomination Award** (Only 3%) 2023
- Excellent Academic Innovation Achievement Award of Beihang University 2024
- Outstanding Graduate Student of Beihang University 2024
- Huawei Scholarship 2024
- First Class Graduate Academic Scholarship 2023
- Graduate Freshman Scholarship 2022
- Three Good Students of Beihang University 2023
- Fengbo Metrology Elite Scholarship 2021
- First Class Scholarship at the School Level (2019, 2020, 2021, 2022) 2019-2022
- Third Prize in China Service Robot Competition 2019

ACADEMIC EXPERIENCE

The Chinese University of Hong Kong (Shenzhen)
Research Assistant

Shenzhen, China
 02/2025 — 07/2025

Eastern Institute of Technology
Research Assistant

Ningbo, China
 05/2024 — 07/2024

SKILLS

- **Programming:** Python/Latex/Matlab
- **Language:** I have successfully passed Ielts exams, published multiple academic papers in English, and possess excellent writing skills in English